

INTRODUCTION TO INFO. & COMM. TECHNOLOGY

20-12-2023

Example-1

Find the
output ?

```
#include <iostream>
using namespace std;
int main()
{
    int num1, num2;
    double sale;
    char first;
    string str;
    num1 = 4;
    cout << "num1 = " << num1 << endl;
    num2 = 4 * 5 - 11;
    cout << "num2 = " << num2 << endl;
    sale = 0.02 * 1000;
    cout << "sale = " << sale << endl;
    first = 'D';
    cout << "first = " << first << endl;
    str = "It is a sunny day.";
    cout << "str = " << str << endl;
    return 0;
}
```

Output - Example-1

```
num1 = 4
num2 = 9
sale = 20
first = D
str = It is a sunny day.

-----
Process exited after 0.07283 seconds with return value 0
Press any key to continue . . .
```

Ex-2

- Write a program in C++ that takes a four digits integer from user and shows the digits on the screen separately i.e. if user enters 5769, it displays 5,7,6,9 separately.

Given code..

The statements with double slash are called comments. Compiler ignore all comments while compiling the source code into object code

```
#include <iostream>
using namespace std;
main()
{
    // declare variables
    int number, digit;
    // prompt the user for input
    cout << "Please enter 4-digit number:";
    cin >> number;
    digit = number % 10;
    cout << "The digits are: ";
    cout << digit << ", ";
    number = number / 10;
    digit = number % 10;
    cout << digit << ", ";
    number = number / 10;
    digit = number % 10;
    cout << digit << ", ";
    number = number / 10;
    digit = number % 10;
    cout << digit;
    return 0;
}
```

Oops.. Can somebody correct the order?

```
Please enter 4-digit number:5769
The digits are: 9, 6, 7, 5
-----
Process exited after 13.69 seconds with return value 0
Press any key to continue . . .
```

This is your Homework

Let's practice Loops now

Ex-1

Write a code to print first 7 integers (1..7) using WHILE loop

Solution

```
#include <iostream>
using namespace std;
int main()
{
    int i;
    i = 1;
    while (i <= 7)
    {
        cout<<i<<endl;
        i = i + 1;
    }
    return 0;
}
```

Ex-2

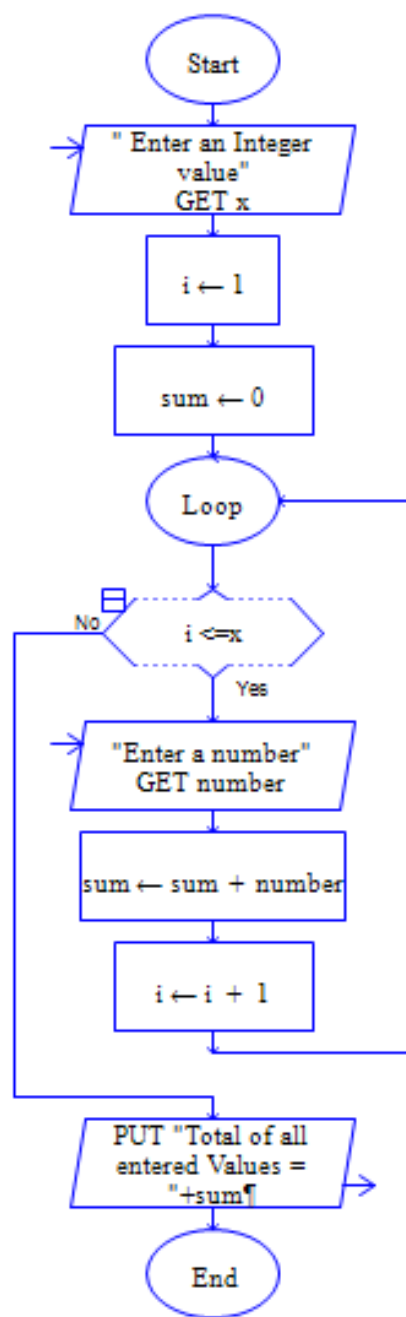
Write a code to print the sum of first 7 integers (1..7) using WHILE loop

Solution

```
#include <iostream>
using namespace std;
int main()
{
    int i;
    int sum;
    sum = 0;
    i = 1;
    while (i <= 7)
    {
        sum = sum + i;
        cout<<i<<endl;
        i = i + 1;
    }
    cout<<endl<<"Total of first 7 integers = "<<sum;
    return 0;
}
```

Ex-3 (attempt using C++)

- Ask the user to enter an integer value, x
- Your loop should run x number of times
- Accept x inputs
- Add all entered numbers
- Display the final sum



Ex-4

Write code to display the square of first five integers using WHILE loop structure:

Square = 1

Square = 4

Square = 9

Square = 16

Square = 25

Ex-5

Write code to display first five integers along with their squares using WHILE loop structure

n = 1	Square = 1
n = 2	Square = 4
n = 3	Square = 9
n = 4	Square = 16
n = 5	Square = 25

Ex - 6

Write a program to print counting from 10 to 1 (in reverse order) using WHILE loop

Ex-7

Ask the user to enter a number. Write code to display the table of the given number in the following format using WHILE loop:

$$2 * 1 = 2$$

$$2 * 2 = 4$$

$$2 * 3 = 6$$

.

.

.

.

Ex-8

- Ask the user to enter marks (out of 100) in ICT of 10 students using WHILE loop and calculate class average score. Display Average score of the class and add a validation check to accept only positive numbers.

Ex-8

- Ask the user to enter marks (out of 100) in ICT of 10 students using WHILE loop and calculate class average score. Display Average score of the class and add a validation check to accept only positive numbers.
- Try float data type for both sum and average and also divide the sum by 10.0 if you are not getting answer with decimal points

Solution - Ex-8

```
{  
    int x, marks;  
    float sum, average;  
    sum = 0.0;  
    average = 0.0;  
    x = 1;  
    while (x <= 10)  
    {  
        cout<<"Enter Marks of the student = ";  
        cin>>marks;  
        if (marks<0)  
        {  
            cout<<endl<<endl<<"Negative Marks cannot be accepted .. Run the Program again.."<<endl;  
            exit;  
        }  
        else  
        {  
            sum = sum + marks;  
        }  
        x = x + 1;  
    }  
    average = sum/10.0;  
    cout<<endl<<"Class Average = "<<average;  
    return 0;  
}
```

Sample Run

```
D:\Fall-2021-Books\ICT\C++-F2021\Lec16-Ex8.exe
Enter Marks of the student = 1
Enter Marks of the student = 2
Enter Marks of the student = 3
Enter Marks of the student = 4
Enter Marks of the student = 5
Enter Marks of the student = 6
Enter Marks of the student = 7
Enter Marks of the student = 8
Enter Marks of the student = 9
Enter Marks of the student = 10

Class Average = 5.5
-----
Process exited after 6.75 seconds with return value 0
Press any key to continue . . .
```

Find the output Ex-9

```
#include <iostream>
using namespace std;
int main()
{
    int d,e,f;
    d = 4;
    e = 6;
    f = 7;
    while (d > f)
    {
        d = d + 1;
        e = e - 1;
    }
    cout <<"Value in d = "<<d<<endl;
    cout <<"Value in e = "<<e<<endl;
    cout <<"Value in f = "<<f;
    return 0;
}
```

```
Value in d = 4
```

```
Value in e = 6
```

```
Value in f = 7
```

```
-----
```

```
Process exited after 0.09106 seconds with return code 0
```

```
Press any key to continue . . .
```

Find the output Ex-10

```
#include <iostream>
using namespace std;
int main()
{
    int g,h,c;
    g = 4;
    h = 9;
    c = 5;
    while (g < h)
    {
        g = g + 1;
        h = h - 2;
    }
    cout <<"Value in g = "<<g<<endl;
    cout <<"Value in h = "<<h<<endl;
    return 0;
}
```


Solution

```
Value in g = 6  
Value in h = 5  
  
-----  
Process exited after 0.1615 second  
Press any key to continue . . .
```

Ex-11

- Write code to print first 10 odd numbers using WHILE loop.

Ex-12

- Ask the user to enter start value and end value. Then generate odd numbers between the entered range (inclusive)

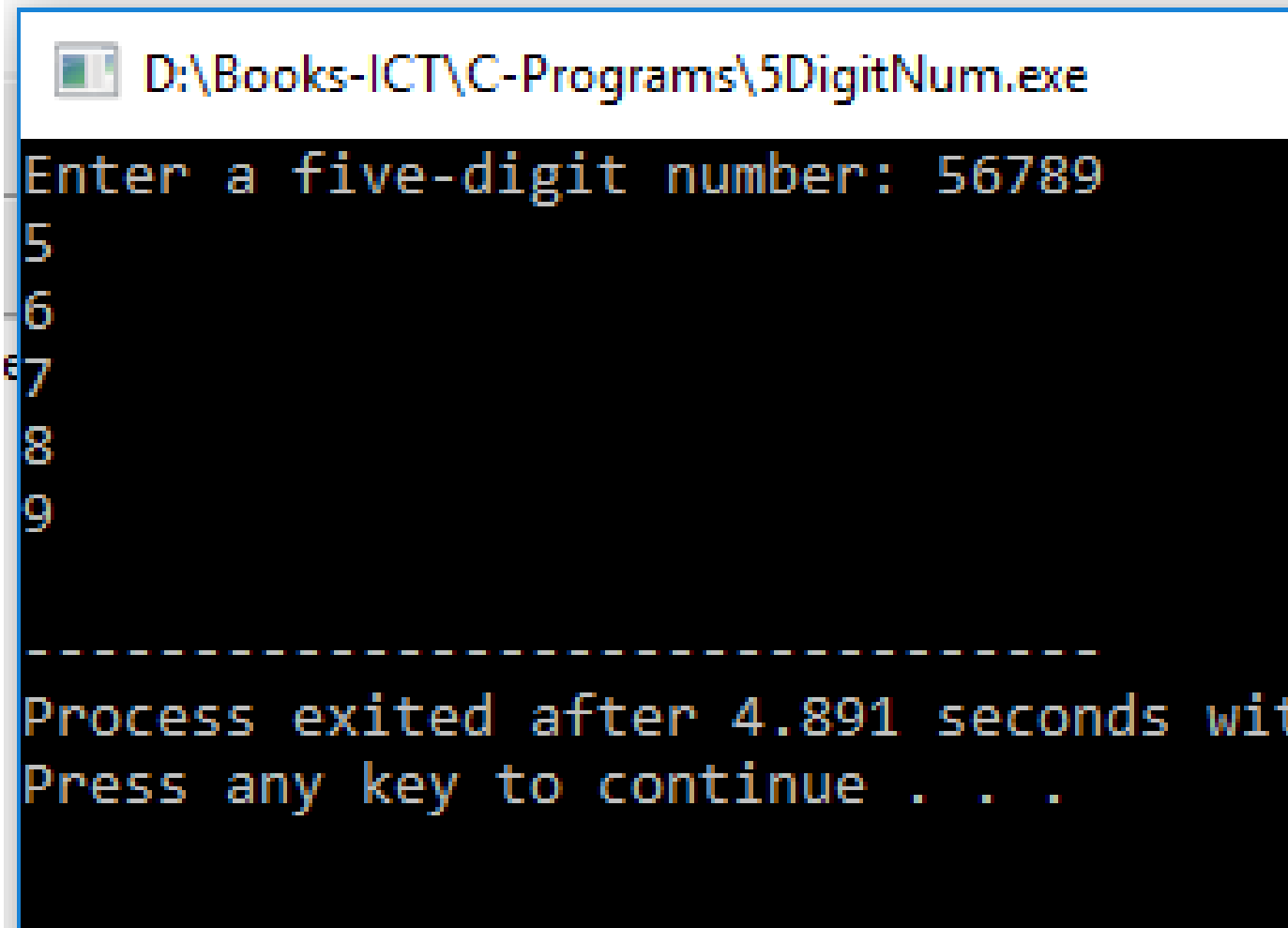
Ex-13

- Ask the user to enter start value and end value. Then find out the largest number between the entered range (inclusive)

Ex-14 (Mod operator: %)

- Ask the user to enter a 5-digit number
- The program should display each digit on a separate line
- Refer to the next slide.

Sample run..



```
D:\Books-ICT\C-Programs\5DigitNum.exe
Enter a five-digit number: 56789
5
6
7
8
9

-----
Process exited after 4.891 seconds with return code 0
Press any key to continue . . .
```

Ref Chapters

- Book 3
- Chp 5